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A deep-learning-based MAC for integrating channel access, rate adaptation, and channel switch ☆

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ABSTRACT

With increasing density and heterogeneity in unlicensed wireless networks, traditional MAC protocols, such as Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA) in Wi-Fi networks, are experiencing performance degradation. This is manifested in increased collisions and extended backoff times, leading to diminished spectrum efficiency and protocol coordination. Addressing these issues, this paper proposes a deep-learning-based MAC paradigm, dubbed DL-MAC, which leverages spectrum data readily available from energy detection modules in wireless devices to achieve the MAC functionalities of channel access, rate adaptation, and channel switch. First, we utilize DL-MAC to realize a joint design of channel access and rate adaptation. Subsequently, we integrate the capability of channel switching into DL-MAC, enhancing its functionality from single-channel to multi-channel operations. Specifically, the DL-MAC protocol incorporates a Deep Neural Network (DNN) for channel selection and a Recurrent Neural Network (RNN) for the joint design of channel access and rate adaptation. We conducted real-world data collection within the 2.4 GHz frequency band to validate the effectiveness of DL-MAC. Experimental results demonstrate that DL-MAC exhibits significantly superior performance compared to traditional algorithms in both single and multi-channel environments, and also outperforms single-function designs. Additionally, the performance of DL-MAC remains robust, unaffected by channel switch overheads within the evaluation range.

1. Introduction

Today, the demand for mobile smart devices and diverse Internet applications is rising, leading to an expected increase in mobile network traffic to 403 exabytes per month by 2029 [1]. With this trend, the number of devices employing short-range communication technologies such as Wi-Fi, Bluetooth, Zigbee, MulteFire, LTE-Unlicensed (LTE-U), and New Radio Unlicensed (NR-U) is rapidly increasing, which results in a coexistence of dense devices in the unlicensed band, posing challenges to user experiences. This shift has intensified the heterogeneity of network environments, making the coordination between different protocols even more complex.

In such a congested and heterogeneous wireless network, traditional Medium Access Control (MAC) protocols are facing severe challenges. These MAC protocols were originally designed for homogeneous networks where all devices follow the same rules. However, in complex and heterogeneous networks, they are inefficient at coordinating conflicts and interference among devices using different protocols, thereby struggling to meet the transmission efficiency and reliability requirements. For example, as observed in [2–4], the Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA) MAC protocol in Wi-Fi networks, does not coexist effectively with other MAC protocols, including even the simplest ones like Time Division Multiple Access (TDMA).

Another issue that deserves attention is that in traditional network protocol stacks, distinct operational functions are often designed as independent modules. The algorithm designed within each module focuses solely on a specific function and lacks collaborative perspectives with other modules. Take rate adaptation at the physical layer and channel access at the MAC layer as examples. Rate adaptation algorithms typically focus on adapting to the path loss of wireless channels, whereas

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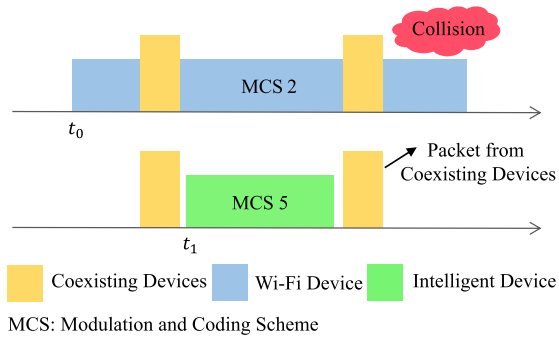


Fig. 1. A toy example of traditional and intelligent transmission protocols.

channel access mechanisms (e.g., CSMA/CA in Wi-Fi) are primarily concerned with collision avoidance. By shifting the paradigm and developing a novel MAC algorithm in a holistic manner that joints channel access and rate adaptation, it can potentially achieve substantial performance improvements.

Fig. 1 illustrates a toy example of a heterogeneous coexisting network, showing the behavioral differences between traditional and intelligent protocols during transmission. As shown in Fig. 1, the blue and green rectangles depict the transmitting packets from 2 devices of interest, each operating under a different protocol, while the orange rectangle represents concurrent transmission from coexisting devices. In the upper diagram, the transmission of a Wi-Fi device, which is represented by the blue rectangle, utilizes the traditional CSMA/CA mechanism for channel access, employing a commonly adopted rate adaptation algorithm such as Intel IWL [5] to dynamically adjust its transmission rate. At time t_0, after the requisite carrier sensing and backoff, the Wi-Fi device decides to transmit using a Modulation and Coding Scheme (MCS[1]) indexed by 2, as determined by the rate adaptation algorithm. We can see that a collision occurs between the Wi-Fi device and the coexisting devices, due to the Wi-Fi device's inability to predict the transmission behaviors of the coexisting devices. In contrast, the lower diagram shows an intelligent device, with its transmission represented by the green rectangle, which at time t_1 leverages its learned insights into the wireless environment to predict the activities of coexisting devices. This prediction capability allows it to make a precise channel access decision, coupled with an appropriate MCS selection, i.e., MCS 5, which avoids collisions and enables a higher transmission rate. As depicted in Fig. 1, the key to designing an intelligent MAC algorithm lies in the ability to discern the transmission behaviors of coexisting devices by observing the wireless environment. This is achieved by employing a state-of-the-art data-driven deep learning framework. Consequently, the MAC protocol is nothing but a neural network that processes a temporal sequence of channel data to determine joint channel access and rate adaptation strategies. The input channel data are readily extracted from the energy detection module common in wireless devices, enabling the neural network to infer whether and at what rate to access the channel based on historical spectrum conditions.

Furthermore, the channel switch mechanism can effectively cope with the dynamic changes in channel quality within crowded and heterogeneous wireless networks. However, traditional channel switch protocols often lack the necessary flexibility and ability to respond rapidly to changes in network conditions. Traditional switching strategies tend to be based on static settings and are incapable of adjusting to real-time changes in network conditions. Such static methods are less effective when the network environment changes rapidly, especially when channels become congested or interference intensifies. Moreover, traditional channel switch algorithms often overlook the communication needs of users and quality of service parameters, which may impose unnecessary delays on sensitive traffic under high network loads. Additionally, strategies involving frequent channel switches lead to intermittent link disruptions and exacerbate performance degradation due to poor decision-making. In this context, a more flexible and dynamic channel switch mechanism would be more suited to the needs of complex and heterogeneous networks.

In response to the challenges of traditional protocols, this paper presents a deep-learning-based MAC paradigm, dubbed DL-MAC, which offers a holistic approach that integrates channel access, rate adaptation, and channel switch into a unified system. Specifically, the DL-MAC protocol is a deep neural network paradigm that takes spectrum sequences freely acquired from wireless devices as input. By utilizing the advanced feature extraction and pattern recognition capabilities of Deep Learning (DL), DL-MAC can extract information from these spectrum sequences, enabling more efficient cross-layer decision-making on channel selection and the joint design of channel access and rate adaptation operating on a specific channel. To the best of our knowledge, we are the first to integrate channel access, rate adaptation, and channel switch. We have validated its performance on a real-world dataset, which has been made publicly available on GitHub for academic and research use.[2]

Overall, the contributions of this work are summarized as follows:

- We propose a deep-learning-based protocol, dubbed DL-MAC. This protocol integrates multiple functionalities, including channel access, rate adaptation, and channel switch, aiming to optimize the spectrum efficiency and enhance the adaptability of wireless networks.
- We have sampled wireless spectrum data from the 2.4 GHz Industrial, Scientific, and Medical (ISM) frequency band in the real world and have made this dataset open-source and available to the GitHub community for public use. We have further verified the performance of the DL-MAC protocol on this dataset. The results show that DL-MAC has superior performance of throughput and lower latency compared to traditional algorithms in both single-channel and multi-channel environments.
- To evaluate the performance gains yielded by the joint functional design of channel access and rate adaptation, we compared it with single-function designs. In particular, these single-function designs involve either a DL-based channel access combined with traditional MCS selection, or traditional CSMA/CA channel access paired with a DL-based rate adaptation strategy. The experimental results show that DL-MAC, when operating on a single channel, significantly improves performance in terms of throughput and mean delay compared to the single-function designs.
- To confirm the protocol efficiency of DL-MAC in multi-channel settings, we assessed the potential performance overheads incurred by channel switch. The experiments demonstrate that the performance of DL-MAC is not significantly affected by channel switch overheads within the tested range, proving its effectiveness in multi-channel settings.

1.1. Related work

In the IEEE 802.11 standard, CSMA/CA serves as the core mechanism for channel access. Early research on the CSMA/CA protocol mainly focused on optimizing parameters and improving algorithms, such as adjusting the size of the contention window [6–8], refining the backoff algorithm [9–11], and enhancing the efficiency of the Request To Send/Clear To Send (RTS/CTS) mechanisms [12,13]. With the advancement of Artificial Intelligence (AI) technology, learning-based

[1] Herein, it is noted that rate adaptation and MCS selection both refer to the task of adjusting transmission rates based on the quality of the wireless channel, and we use these terms interchangeably throughout this paper.

[2] The dataset is available at https://github.com/postman511/Wireless-Signal-Strength-on-2.4GHz-WSS24-dataset.

optimization strategies have also been proposed for these parameters and the enhancement of algorithms [14–17]. Furthermore, [18] introduced a new method for designing network protocols based on Deep Reinforcement Learning (DRL), where DRL agents can selectively combine different functionalities to enhance the performance of the protocol. However, this approach is only targeted at the CSMA protocol and has yet to provide a solution for the issues between different MAC protocols in heterogeneous networks. Note that, all these improvements to traditional algorithms, as well as learning-based methods, primarily focus on enhancing node throughput and access fairness, and they do not sufficiently address the issue of coexisting devices using different protocols in complex and heterogeneous networks. The authors in [3] and [19] utilized DRL to design multi-access techniques tailored for heterogeneous networks, aiming to maximize throughput and ensure fairness in access while coexisting with other protocols. However, these studies based on simulated data have not validated the performance of the proposed protocols in realistic scenarios and lack consideration in terms of rate adaptation.

Beyond the MAC layer's channel access mechanisms, Rate Adaptation Algorithms (RAA) at the physical layer have also been extensively studied. The ARF [20] algorithm dynamically adjusts the data transmission rate based on consecutive transmission successes, increasing or decreasing the rate accordingly, while SampleRate [21] and Minstrel [22] schemes optimize rate selection based on past performance and statistics. Additionally, the Intel IWL [5] algorithm also dynamically adjusts rates but it specifically uses historical throughput and transmission count data to inform its adjustments. However, these algorithms aim to predict and adjust to the optimal future transmission rate based on past network performance data, employing certain exploration strategies to find the suboptimal rate. Recently, several studies have adopted AI technologies to achieve more accurate predictions and better adaptability in response to changing network conditions. The authors in [23] used a three-layer fully connected network to predict the best rate and adjust it according to real-time network conditions. [24] utilized deep Q-learning to adapt to current link quality and channel conditions, implementing an intelligent rate adaptation algorithm. Apart from the above, the authors in [25] proposed a DRL-based MCS selection algorithm in heterogeneous networks. However, all these algorithms solely focus on enhancing the performance of their specific functions of rate adaptation, without taking into account other functionalities, such as channel access.

What's more, dynamic channel switching is also a key research topic for enhancing wireless network performance, especially in high-density environments. Early studies mainly focused on static channel allocation strategies, which often rely on the fixed attributes of channels, such as signal strengths or interference levels [26,27]. However, these methods perform poorly under dynamically changing network conditions. As demands for network performance have increased, researchers have begun to shift towards dynamic channel selection that considers real-time network status. Representative works in this area include adaptive methods based on channel occupancy [28], network congestion [29], and channel interference [30]. The introduction of AI techniques has opened up new possibilities for dynamic channel selection. The authors in [31] employed machine learning to identify bad neighbors in Wi-Fi networks and allocate the optimal channel for each network to maximize the overall network performance. In [32], the authors proposed a DL-based channel allocation scheme aimed at minimizing co-channel interference (CCI) among Access Points (APs), by utilizing a DNN to predict the channel allocation of each AP to minimize CCI. Although data-driven channel selection strategies can optimize channel selection decisions based on real-time network conditions and historical data, current research results mainly rely on simulation environments and have not been widely validated in real-world environments to prove their performance. This means that while these strategies perform well in theory and simulated tests, their actual effectiveness and reliability under the complex and constantly changing real-world conditions remain to be further explored and confirmed.

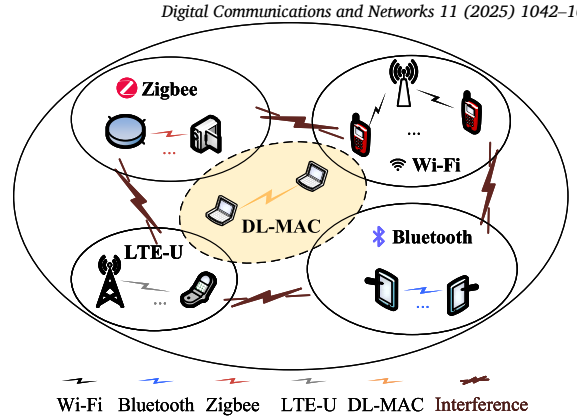


Fig. 2. Coexistence of devices utilizing various protocols within the 2.4 GHz frequency band.

In our previous work [33], we developed a DL-MAC protocol that focused on the joint design of channel access and rate adaptation on a specific channel to maximize spectrum efficiency. Differing from [33], this paper extends DL-MAC to operate channel access from one channel to multiple channels, i.e., endowing DL-MAC with the ability of channel switching. In addition, we evaluated the effectiveness of the protocol in terms of throughput but did not validate its performance in terms of packet transmission delay in [33]. In this work, DL-MAC is designed to enhance AI devices to operate in a multi-channel environment, by integrating channel switch functionality to adaptively select the most suitable channel for transmission. Moreover, we have validated the proposed protocol, with a focus on its performance in terms of both throughput and mean delay.

1.2. Outline

The remainder of this paper is organized as follows: Section 2 presents the system scenario and preliminaries. Section 3 describes the raw dataset, followed by Section 4, which details the specific implementation methods of the DL-MAC protocol. Numerical experiments conducted to verify the proposed DL-MAC are discussed in Section 5. Finally, in Section 6, we conclude our work and discuss some possible research directions in the future.

2. Background and preliminary

In this section, we will provide background information on the system scenario and preliminaries in Wi-Fi networks.

2.1. System scenario

We consider a Wi-Fi network operating on the 2.4 GHz ISM frequency band. This network comprises both legacy Wi-Fi devices and AI devices, adhering to the CSMA/CA MAC protocol and the proposed DL-MAC protocol, respectively. In addition to Wi-Fi systems, devices operating under disparate wireless protocols such as Bluetooth, Zigbee, and LTE-U commonly coexist within the 2.4 GHz frequency band, as illustrated in Fig. 2. These devices operate independently, each abiding by their respective protocols. Consequently, the concurrent operation of these diverse networks precipitates mutual interference, resulting in considerable spectrum inefficiency and deteriorations of Wi-Fi communication quality.

2.2. CSMA/CA mechanism

The Listen-Before-Talk (LBT) principle is fundamental to the CSMA/CA protocol, aimed at reducing packet transmission conflicts in shared channels. Under LBT, devices must sense that the channel is idle before

initiating packet transmission. Additionally, the protocol incorporates a random backoff mechanism that introduces delays before transmission, further reducing the chances of collisions. A device can transmit only after it has consistently sensed that the channel has been idle for a duration of the Distributed Coordination Function Inter-frame Spacing (DIFS). After this DIFS period, the device enters a random backoff period, selecting a backoff counter and waiting for that duration, while pausing the backoff counter if the channel becomes busy. Once the backoff counter reaches zero, the device gains channel access for packet transmission. CSMA/CA employs a binary exponential backoff algorithm, where the backoff counter is uniformly chosen from (0, CW-1), with CW representing the contention window. Upon initial transmission failure, CW begins at CW_{min} and doubles up to CW_{max}, calculated as $2^P CW_{min}$, with P being the maximum backoff exponent to limit transmission attempts.

In dense network environments, CSMA/CA is more prone to transmission collisions, leading to extended backoff times for each device. The timing of transmission is heavily influenced by the outcome (success or not) of the previous ones. This inspired us to design a specialized channel access protocol for independent channel access decision-making.

2.3. Rate adaptation

In licensed bands, prevalent RAAs rely on Signal-to-Noise Ratios (SNR). That is, the transmitter determines the MCS based on the SNR feedback or estimates from the inverse link. However, SNR-based methods often underperform in the high-interference unlicensed band because the optimal MCS depends not only on noise but also on interference from coexisting devices. In practice, most RAAs used in the unlicensed band employ sampling-based approaches. These algorithms sample statistical data, such as Packet Error Rates (PER) and throughput across different rates, to evaluate channel conditions. Subsequently, they select the rate that has historically performed the best and employ it as the current optimal transmission rate. However, this approach can sacrifice some performance since it occasionally explores suboptimal rates to gather enough statistics. This motivates us to implement a more efficient RAA that directly selects rate options based on various channel conditions, without the necessity of exploring suboptimal rates.

2.4. Channel switch

Channel switching dynamically adapts to changing channel conditions, aiming to reduce interference and enhance network performance. Typically, an AP within a basic service set scans for and assesses the channel quality and congestion levels of the surrounding channels. Based on this assessment, the AP selects the optimal channel to communicate with the devices it manages. However, in complex and dense networks, frequent channel switches due to noise and congestion can consume substantial time and resources, impacting user experience and network stability. Often, these switches are based on immediate signal quality assessments, which may overlook future channel uncertainties. Such a short-term focused approach can lead to decision biases and errors. This challenge inspired us to develop an intelligent channel switching strategy to effectively predict channel conditions under varying network conditions, thereby reducing frequent switching and avoiding biases and errors arising from short-term decision-making.

3. Dataset

Our proposed DL-MAC utilizes real-world wireless spectrum data collected from the 2.4 GHz ISM frequency band. This section briefly introduces the dataset and details the data processing procedure.

3.1. Raw data collection

The Signal-to-Interference-plus-Noise Ratio (SINR) is crucial for assessing wireless channel quality, influencing channel access, and MCS selection. Accordingly, we propose using the Received Signal Strength Indicator (RSSI) as a proxy for SINR, exploiting its easy acquisition from energy detection modules in wireless devices for channel sensing. This approach simplifies the process of assessing wireless channel quality, leveraging RSSI's availability and ease of measurement.

For real-world RSSI data, we measured their values in the 2.4 GHz ISM band using the Ellisys Bluetooth Vanguard (BV1), capable of capturing radio frequency spectrum data across 79 channels, each with a bandwidth of 1 MHz within the 2.4 GHz band. This includes signals from Bluetooth and other devices such as Wi-Fi, LTE-U, Zigbee, and ANT. The BV1's real-time data collection from various networks offers comprehensive insights into the spectrum usage in this band.

We define each 1-MHz sub-band as $f_n = 2402 + n$ (MHz), where $n = 0, 1, \cdots, N$ and the maximum value of N is 78. RSSI values are sampled from each sub-band at a consistent interval T_s, set to 100 μs. The sampling period comprises L RSSI samples. For N sub-bands, we periodically compile these RSSI samples into a comprehensive matrix as shown below:

$$\boldsymbol{RSSI} = \begin{bmatrix} RSSI_{t_1,f_0}, & \cdots, & RSSI_{t_1,f_N} \\ \vdots & \vdots & \vdots \\ RSSI_{t_L,f_0}, & \cdots, & RSSI_{t_L,f_N} \end{bmatrix} \tag{1}$$

In this matrix, each row represents the sampled data across different sub-bands at a single sampling point in time, while each column captures the sampled data over time for the same sub-band. Thereby, $RSSI_{t_l,f_n}$ represents the (l, n)-th entry, where $t_l = t_0 + T_s l$. Here, t_0 is the starting time, and l represents the l-th sample, ranging from 1 to L. This raw RSSI matrix encapsulates the entire dataset, representing the spectrum of RSSI values across all the observed sub-bands.

Note that L denotes the size of the sample set collected in our dataset. Due to the fine-grained sampling interval of 100 μs, the dataset is significantly large. Without loss of generality, in subsequent sections, we focus our design and experiments on data from the first two minutes of this dataset to validate the performance of our proposed DL-MAC protocol. It is noteworthy that the raw dataset is open-source and available for public use.

3.2. Data preprocessing

In Wi-Fi networks, the mini-slot, T_{slot}, is the basic time unit, set to 9 μs [34]. Thus, we process the raw RSSI data from the 79 sub-bands to align with Wi-Fi standards and convert it to fit the 13 Wi-Fi network channels.

Initially, we align the raw RSSI data with Wi-Fi network time slots in the time domain. The sampling interval of RSSI data is $T_s = 100$ μs, differing from the mini-slot, T_{slot}, in Wi-Fi networks. To align the time units of the CSMA/CA protocol with the RSSI data, we use linear interpolation for up-sampling with a factor of $T_s/T_{slot} \approx 11$. Specifically, for an interpolation point t', we identify the two closest time points t_1 and t_2 around it and use the original values at these points for linear interpolation. Assume that $RSSI_{i,f_j}$ represents the elements at the i-th row and j-th column in the original matrix. Then, for interpolation point t' on sub-band f_j, the interpolated $RSSI_{t',f_j}$ can be calculated using the following formula:

$$RSSI'_{t',f_j} = \frac{RSSI_{t_2,f_j} - RSSI_{t_1,f_j}}{t_2 - t_1}\left(t' - t_1\right) + RSSI_{t_1,f_j} \tag{2}$$

where t_1 and t_2 are the nearest time points to t', located immediately to the left and right, respectively.

After time domain processing, we map the data to Wi-Fi network channels. Wi-Fi networks typically have 13 overlapping channels, with the central frequency of the i-th channel defined as $F_i = 2412 + 5(i-1)$ MHz, for i ranging from 1 to 13. We use average interpolation with a down-sampling factor of 21, corresponding to the bandwidth of a Wi-Fi channel. For instance, channel 6, covering 21 sub-bands from 2.427 GHz

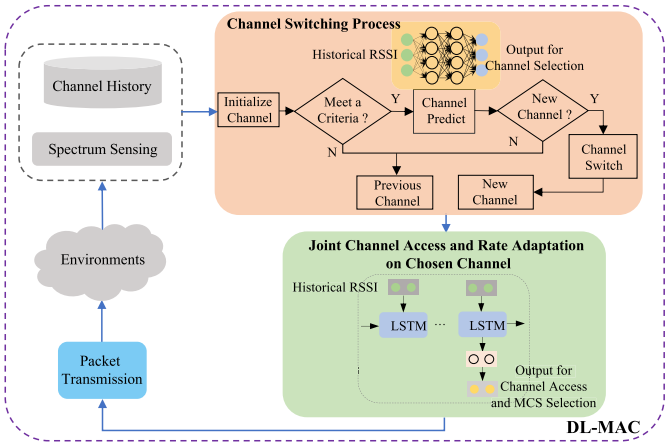


Fig. 3. Overview of DL-MAC framework.

to 2.447 GHz ($f_{25}, \cdots, f_{45}$), averages these sub-bands' RSSI data for its value. Mathematically, the RSSI for the i-th Wi-Fi channel at time t'_l is $RSSI'_{t'_l,i} = \frac{1}{21}\sum_{f=F_i-10}^{f=F_i+10} RSSI'_{t'_l,f}$. Consequently, the final RSSI matrix can be obtained as follows:

$$\boldsymbol{RSSI'} = \begin{bmatrix} RSSI'_{t'_1,1}, & \cdots, & RSSI'_{t'_1,13} \\ \vdots & \vdots & \vdots \\ RSSI'_{t'_{L'},1}, & \cdots, & RSSI'_{t'_{L'},13} \end{bmatrix} \tag{3}$$

Note that, the original RSSI matrix dimensions are $L \times N$, where L is the number of samples, and N represents the number of sub-bands at each sampling time point. Through the interpolation process in the time domain, the number of samples increases to $L' = L \times \lfloor T_s/T_{slot} \rfloor$. Additionally, interpolation in the frequency domain modifies the number of columns in the original RSSI matrix. Initially, the matrix had N columns corresponding to the original sub-bands; after interpolation, this scales to 13 columns, each representing the RSSI values for one of the 13 channels. Consequently, the dimensions of the interpolated RSSI matrix become $L' \times 13$. Generally, if we focus on a particular channel, such as channel i, the RSSI matrix can be reduced to a vector, denoted as $\boldsymbol{RSSI'} = [RSSI'_{t'_1,i}, \cdots, RSSI'_{t'_{L'},i}]T$. For clarity in subsequent discussions, we will uniformly refer to the processed data as RSSI in the following.

4. DL-MAC protocol

In this section, we will first provide an overview of our DL-MAC framework design, then detail the joint channel access and rate adaptation mechanism for a specific channel, and finally, describe our approach for implementing channel switching.

4.1. DL-MAC framework

Fig. 3 details the DL-MAC framework, emphasizing its 2 main components: channel switching and the joint design of channel access and rate adaptation on the selected channel. Unlike traditional methods, DL-MAC relies on sequential channel sensing results and employs DL for decision-making. It integrates a DNN for channel selection decisions and an RNN for joint channel access and rate adaptation on a selected channel. A fundamental requirement for DL-MAC is its capability to continuously sense all available channels in real-time, with the data stored in a constantly updated channel history database. This database provides the input for the DNN and RNN to make decisions. When channel switching conditions are met, the DNN activates and determines the optimal channel for subsequent transmission. If the chosen channel matches previous selections, DL-MAC remains on the current channel; otherwise, it initiates switching. After switching, the RNN uses the historical data of the selected channel to perform channel access and rate adaptation jointly. It should be noted that, when DL-MAC is designated to operate on a specific channel, it functions for single-channel use and solely focuses on the joint design of channel access and rate adaptation, as delineated in our earlier work [33].

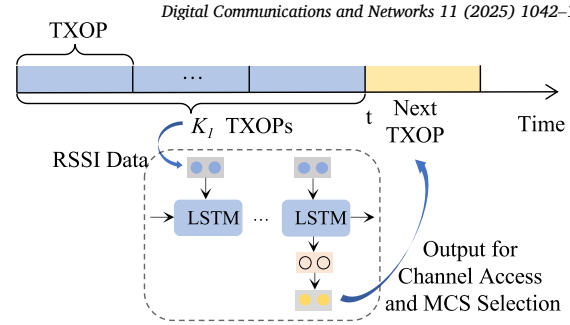


Fig. 4. Data labeling and RNN.

Table 1
MCS indices and corresponding minimum SINR required.

MCS Index	Modulation	Coding	Rate (Mbps)	Min. SINR (dB)
-1	-	-	-	<2
0	BPSK	1/2	6.5	2
1	QPSK	1/2	13	5
2	QPSK	3/4	19.5	9
3	16-QAM	1/2	26	11
4	16-QAM	3/4	39	15
5	64-QAM	2/3	52	18
6	64-QAM	3/4	58.5	20
7	64-QAM	5/6	65	25
8	256-QAM	3/4	78	28

4.2. Joint channel access and rate adaptation

In this subsection, we explore the mechanism of joint channel access and rate adaptation for a specific channel, beginning with the essential data labeling process. Next, we detail the structure of the neural network involved, followed by outlining the specific implementation steps to give a comprehensive view of the procedure.

4.2.1. Data labeling procedure

In our study, channel access and MCS selection are performed using supervised learning, where dataset labeling is a crucial initial step.

We define a Transmit Opportunity (TXOP[3]) spanning 120 mini-slots, and Fig. 4 illustrates the RNN alongside a training sample. At time t, we use historical RSSI data from the past K_1 TXOPs as input. The input data comprises K_1 sequences of 120 RSSI values each. The input's label indicates the optimal channel access and MCS selection for the next TXOP (from $t+1$ to $t+120$), specifically the MCS from Table 1 that maximizes the transmission rate without exceeding a target PER. As we all know, PER is dependent on SINR and the chosen MCS_i, expressed as $PER = f\left(SINR, MCS_i\right)$. With a constant SINR, PER increases monotonically with MCS_i. For each MCS_i, to maintain a PER below a target T_{PER}, we calculate the minimum SINR ($SINR_i^{min}$) that meets $f\left(SINR_i^{min}, MCS_i\right) = T_{PER}$. This allows us to establish the operational SINR range for each MCS_i as $[SINR_i^{min}, SINR_{i+1}^{min})$, where MCS_i attains the maximum transmission rate within the target PER. Table 1 enumerates the minimum SINR requirements corresponding to various MCS indices for operational purposes [35]. We also outline a detailed workflow for the data labeling process as follows:

[3] A Transmit Opportunity (TXOP) is a designated period when a Wi-Fi device can access the channel for data transmission. It sets the maximum time a device can occupy the channel, defined here in terms of the number of mini-slots.

- Calculate the average RSSI: For each time t, we gather RSSI data $D^t_{K1} = \{RSSI_{t-120K_1+1}, \cdots, RSSI_t\}$ from the previous K_1 TXOPs, which span $120K_1$ mini-slots. Then, we observe the RSSI values for the upcoming TXOP time intervals, i.e., $\{RSSI_{t+1}, \cdots, RSSI_{t+120}\}$, and calculate their average RSSI, given by $\overline{RSSI} = \frac{1}{120}\sum_{l=1}^{l=120} RSSI_{t+l}$.
- Calculate SINR: We calculate the average SINR at the receiver by applying $SINR = P_r - \overline{RSSI}$. In this calculation, SINR is calculated as the difference in decibels between P_r and $\overline{RSSI}$, where P_r denotes the received signal power at the receiver. In practical systems, P_r can be easily estimated at the transmitter. For our experiments, we assume P_r to be -65 dBm for simplicity, without loss of generality.
- Lookup MCS: We determine the MCS for the upcoming TXOP based on the calculated SINR and the minimum SINR in Table 1. For example, if the calculated SINR is 12 dB, D^t_{K1} is labeled as MCS_3.

4.2.2. Neural network design

The RNN used for joint channel access and rate adaptation includes an input layer, 2 hidden layers, and an output layer. The input layer has 120 neurons, corresponding to the RSSI data from the TXOP. The 2 hidden layers consist of a Long Short-Term Memory (LSTM) layer and a feedforward layer. The output layer generates a probability distribution over MCS indices using a softmax layer. This output enables the prediction of joint channel access and MCS selection at time t, supporting the DL-MAC's operation on a specific channel.

4.2.3. Implementation of joint channel access and rate adaptation

After labeling, we train the RNN using the Adam algorithm [36] and cross-entropy loss on the labeled RSSI data. The trained model is then used for joint channel access and rate adaptation.

Initially, the AI device starts with an idle state and listens to the channel. At each mini-slot t, it collects the RSSI data and then appends it to the tail of the RSSI queue. Subsequently, it inputs the latest $120K_1$ RSSI data into the RNN to generate an MCS_i. In cases where MCS_i equals -1 (indicating no channel access), the AI device continues listening to the channel and acquiring new RSSI data at the next mini-slot $t+1$. If the RNN output indicates transmission, the device transmits the earliest packet in its buffer using MCS_i (not equal to -1) in the upcoming 120 mini-slots. After transmission, it receives ACK or NACK feedback. During transmission, due to half-duplex constraints, the device cannot receive any signal from the wireless channel. Therefore, it becomes necessary to generate RSSI data in a handcrafted manner for this packet transmission time after the transmission is finished. This data, reflecting interference levels based on transmission success or failure, is added to the queue for subsequent RNN predictions.

When MCS_i is ready for packet transmission; if a packet is successfully transmitted, it is assumed that interference from other devices is low. Consequently, the RSSI data is generated uniformly within the range from the RSSI value corresponding to $SINR_i^{min}$ to the RSSI value corresponding to $SINR_8^{min}$. Conversely, if packet transmission fails, this suggests high interference from other devices. In this case, the RSSI data is uniformly generated within the range from the RSSI value corresponding to $SINR_{-1}^{min}$ to the RSSI value corresponding to $SINR_{i-1}^{min}$.

4.3. Chanel switch

The channel switching mechanism, leveraging SINR metrics, enables AI devices to assess all accessible channels and determine the most suitable one guided by a DNN that predicts future network conditions. Upon choosing the optimal channel, AI devices perform joint channel access and rate adaptation for it, allowing adaptation to dynamic network conditions. This subsection outlines the data labeling process, neural networks, and workflow involved in channel switching.

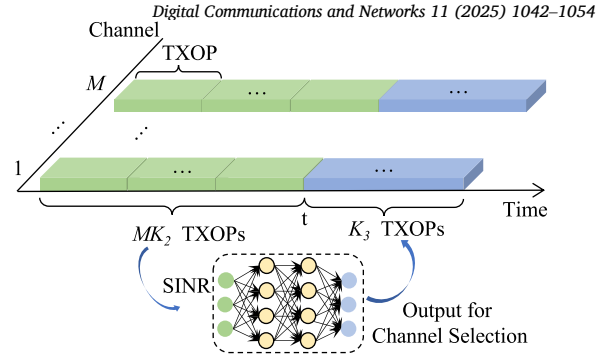


Fig. 5. Data labeling and DNN.

4.3.1. Data labeling process

Similar to the joint design of channel access and rate adaptation, channel switching is also executed within a supervised learning framework. Consequently, it is essential to label the RSSI data for channel switching. Likewise, each TXOP consists of 120 mini-slots.

Assuming AI devices can dynamically switch among the M available channels, as shown in Fig. 5. We use historical RSSI data from all M available channels to predict the channel qualities in the near future, and select the channel with the highest predicted quality for channel access and rate adaptation in the next period, i.e., the duration of $K3$ TXOPs. Specifically, for each channel within the M channels, we collect RSSI data from the past K_2 TXOP intervals and observe RSSI data for the upcoming K_3 TXOP periods. For the m-th channel, such data is denoted as $D^t_{m,K2} = \{RSSI_{m,t-120K_2+1}, \cdots, RSSI_{m,t}\}$ for past intervals and $D^t_{m,K3} = \{RSSI_{m,t+1}, \cdots, RSSI_{m,t+120K_3}\}$ for upcoming intervals, respectively. Subsequently, the average SINR values for these datasets, denoted as $SINR^t_{m,K_2}$ and $SINR^t_{m,K_3}$, are calculated in accordance with the methodology described in Section 4.2. Consequently, the feature set for a sample at time t is $SINR^t_{K_2} = \{SINR^t_{m,K_2}\}^M_{m=1}$, and its corresponding label is the channel from these M options with the highest average $SINR^t_{m,K_3}$ in the upcoming K_3 TXOP periods. This effectively identifies the optimal channel for the specified time intervals.

4.3.2. Neural network design

The DNN designed for channel switching adopts a fully connected architecture consisting of 5 layers: an input layer, 3 hidden layers, and an output layer. The input layer is designed to handle a feature vector of dimension MK_2, where M represents the number of accessible channels and K_2 denotes the quantity of independent SINR values per channel. Following each hidden layer, a Rectified Linear Unit (ReLU) activation function is employed. The network concludes with an output layer, which utilizes a softmax activation function to generate a probability distribution for each channel.

4.3.3. Implementation of channel switching

After the labeling process, we train our DNN using the stochastic gradient descent algorithm and the cross-entropy loss function. Upon training completion, the model is deployed onto AI devices.

Initially, an AI device sets a time interval T_c for channel switching and selects a channel ch_i at random for joint channel access and rate adaptation, as elaborated in Section 4.2.3. Note that, the AI device maintains individual RSSI queues for each channel, appending collected RSSI data to respective queues at every mini-slot. Additionally, the AI device initiates a timer based on a preset interval. When this timer reaches the predetermined T_c and the device is idle, the channel switching process is initiated; otherwise, it maintains its operations on the current channel. Upon entering the channel switching phase, which involves transitioning from one channel to another, the AI device performs a series of configurations and adjustments, including modulating its receiving and transmitting frequencies to match the new channel. During

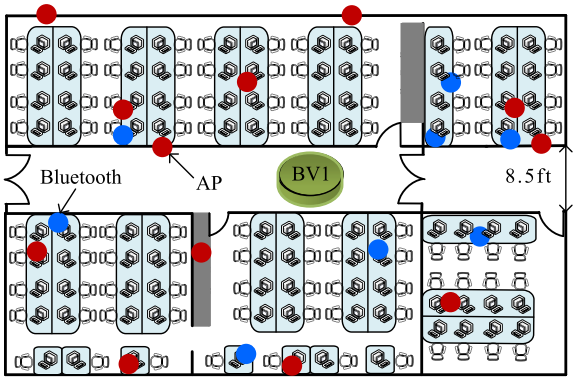


Fig. 6. A schematic layout of wireless data sampling in the laboratory environment at Shenzhen University.

this process, the device is temporarily unable to receive signals from any channel. It resumes signal reception only after completing these adjustments and fully transitioning to the new channel. For simplicity, our primary focus is on the channel selection aspect. Other related configuration activities during the channel switch are collectively determined by time T_d, known as the channel switching overheads.

The AI device selects channel ch_j with the highest probability as the newly selected channel. If ch_j differs from the previously chosen channel ch_i, a switch occurs, followed by joint channel access and rate adaptation on ch_j. Otherwise, the device continues operating on the current channel ch_i. Due to half-duplex constraints, the AI device cannot receive signals and RSSI compensation is necessary for all channels during the transmission and switching phases. For the selected channel, RSSI compensation follows the procedures outlined in the implementation of joint channel access and rate adaptation. For other channels, compensation values are randomly chosen from the entire RSSI range.

5. Experimental results

This section outlines experiments that assess the performance of DL-MAC. We will detail the experimental setup, performance metrics, comparison algorithms, and finally present the results.

5.1. Experimental settings

5.1.1. Sampling environment

Our data sampling was conducted in a laboratory at Shenzhen University where we established an environment with 12 Wi-Fi APs and 8 Bluetooth devices for data collection. The positions of these APs and Bluetooth devices were randomly assigned, with our data sampling hardware, BV1, strategically placed at the center. The layout of this wireless sampling setup is depicted in Fig. 6.

5.1.2. Platform and tools

For the evaluation of DL-MAC and comparative benchmarks, we developed a simulator using Python. Our simulator ran on a computer with an Intel Core i9-10900K CPU and an NVIDIA GeForce RTX 3090 Founders Edition GPU. For neural network training purposes, PyTorch [37] was selected as our DL platform.

5.1.3. System parameters

The fundamental time unit in our experiments is the mini-slot, set to a duration of 9 μs, aligning with Wi-Fi network standards for compatibility. Regarding traffic generation, packet arrivals at the transmitting buffer follow a Poisson distribution, with arrival rate λ (indicating the average number of packets arriving per mini-slot) set to 0.18. Each packet is configured with a default payload size of 1500 bytes and the buffer is designed to accommodate up to 10 packets. Additionally, we have defined 3 distinct values for TXOPs: $K1$, $K2$, and $K3$, set to 3, 5, and 2, respectively.

In the configuration of the CSMA/CA mechanism, the interference threshold is established at -75 dBm. The DIFS is designated a duration of 36 μs. Furthermore, the contention window parameters are CW_{min} set to 32 and CW_{max} to 1024.

5.2. Performance metrics

In this experiment, we assess the performance of DL-MAC, focusing on its ability to maximize throughput while minimizing packet delay. The following metrics are defined to quantify its performance:

- Throughput: This is quantified as the number of successfully transmitted bits per mini-slot, providing a measure of the data transmission efficiency of the system.
- Packet Delay: Defined as the number of mini-slots that elapse from the moment a packet is generated until it is successfully transmitted, this metric measures the latency in packet delivery.
- Mean Delay: Representing the average delay experienced by all successfully transmitted packets, the mean delay provides an overall assessment of the system latency.

5.3. Algorithms for comparison

To evaluate the performance of DL-MAC, we employ several comparative approaches. The optimal transmission policy, derived from the real-world RSSI data, serves as the upper bound for MAC performance. This policy determines the optimum for joint channel access and MCS selection on a given channel. Additionally, we use the traditional CSMA/CA protocol and rate adaptation algorithms as benchmarks. Moreover, to demonstrate the advantages of our joint design, we compare individual DL-based methods for channel access and MCS selection against our joint design. The specific algorithms and combinations used in our benchmarking are as follows:

- OPT: The optimal transmission scheme determines the best decisions for the joint design of channel access and MCS selection on a specific channel. We consider OPT as the upper bound for MAC performance operating on a single channel where DL-MAC operates on.
- CSMA/CA+IWL: This setup uses the CSMA/CA protocol for channel access, combined with the Intel IWL algorithm for MCS selection.
- CSMA/CA+ARF: In this configuration, CSMA/CA is used for channel access, while the ARF algorithm is employed for MCS selection.
- CSMA/CA+DL/MCS: This approach pairs CSMA/CA for channel access with a DL-based algorithm for MCS selection.
- DL/CA+IWL: Here, a DL-based algorithm is applied for channel access, and the Intel IWL algorithm is used for MCS selection.

It should be noted that both CSMA/CA+DL/MCS and DL/CA+IWL are subject to the half-duplex constraint of radio hardware, which prevents AI devices from sensing the channel while transmitting. Consequently, it is necessary to conduct RSSI compensation operations after the transmission has finished. To ensure a fair and accurate comparison with our proposed DL-MAC algorithm, both strategies implement RSSI compensation in accordance with the same principles outlined in subsection 4.2.3.

Additionally, the RTS/CTS control frames serve as an effective channel reservation mechanism that can significantly enhance the performance of Wi-Fi networks. Because RTS/CTS frames are short, reducing the likelihood of collisions, even when collisions occur, entails minimal costs. However, it is important to clarify that our approach, which is based on RSSI data for channel access, differs in its objectives from the RTS/CTS mechanism. Our method aims to avoid potential conflicts with existing devices in the network by accessing idle channels, whereas

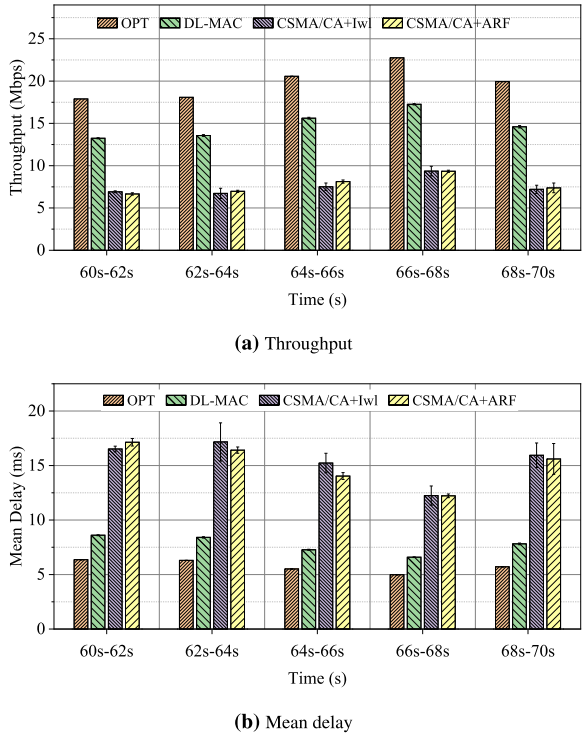


Fig. 7. Performance comparisons of DL-MAC operating on channel 6: (a) Throughput; (b) Mean delay.

RTS/CTS frames are primarily designed to address conflicts during transmission. In wireless networks, due to the need for flexible utilization of unoccupied channels, the traditional RTS/CTS frame control mechanism is not entirely suitable, as it may interfere with existing devices. Therefore, we do not adopt the RTS/CTS mechanism as a baseline for comparison.

5.4. Performance evaluation on a single channel

In experiments, we designate channel 6 as the operational channel for DL-MAC. Initially, we employ offline training for the RNN, utilizing data collected from the first 60 seconds on this channel. The dataset is divided into 2 subsets: 80% is allocated as the training set, and the remaining 20% forms the validation set. To mitigate the risk of overfitting, we employ early stopping, a technique that halts training when there is no observed improvement in the model's performance on the validation set over a specified number of epochs. Following this offline training phase, we conduct online testing to assess the performance of DL-MAC using fresh data from the subsequent 10 seconds. Each experiment is conducted 10 times, with the mean and standard deviation of outcomes reported. Note that the training process is conducted in an offline manner and the testing process is conducted online.

5.4.1. Performance comparison of algorithms

We first compare the performance of DL-MAC against that of traditional algorithms in terms of throughput and mean delay, using OPT as the benchmark for the upper bound of MAC performance. The comparative results for throughput and mean delay are depicted in Fig. 7. Among the protocols compared, OPT consistently outperforms others, achieving the highest throughput and the lowest delay across all time intervals. The traditional algorithms, specifically CSMA/CA+IWL and CSMA/CA+ARF, not only exhibit lower throughput and higher packet delays but also demonstrate considerable performance fluctuations over time. In contrast, DL-MAC achieves significantly higher throughput than the traditional algorithms, as shown in Fig. 7a. In terms of delay performance, Fig. 7b demonstrates that while DL-MAC incurs a mean delay slightly higher than that of OPT, it is substantially lower than those experienced with the traditional algorithms. Taken together, these results show that the performance of DL-MAC, in terms of both throughput and delay, not only outperforms the traditional algorithms but also demonstrates stability and minimal fluctuations, closely approaching the optimal performance benchmark.

5.4.2. Generalization performance

Fig. 8 displays the generalization performance of DL-MAC, which was trained on channel 6 and then tested on channels 1 and 11. The figure also presents 2 performance metrics, i.e., throughput and mean delay, compared against the OPT benchmark and CSMA/CA+IWL on each respective channel.

Across both tested channels, DL-MAC demonstrates better generalization performance than CSMA/CA+IWL, as indicated by higher throughput and lower mean delay. The performance of DL-MAC is closer to that of OPT, proving the effectiveness of DL-MAC in adapting to different channel conditions. Moreover, the traditional CSMA/CA+IWL displays performance fluctuations across various time intervals on both channels 1 and 11, echoing the results previously observed on channel 6. In summary, Fig. 8 demonstrates that DL-MAC has strong generalization capabilities across channels 1 and channel 11, approaching the optimal and surpassing traditional algorithms. This consistent effectiveness, despite the dynamic nature of network conditions, highlights the model's reliable performance and its superiority in handling changes in channel characteristics.

5.4.3. Joint design versus individual design

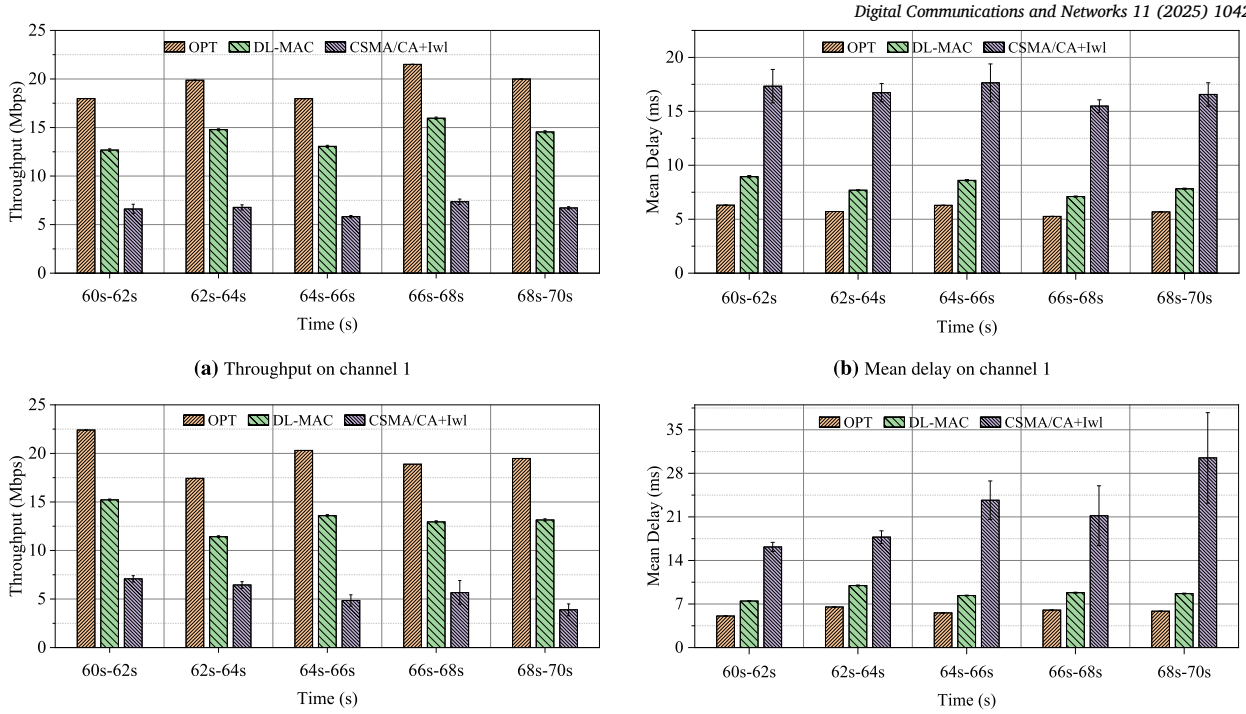
Next, we evaluate the performance of our proposed joint functional design (i.e., the joint design of channel access and rate adaptation) against 2 approaches employing separate functional designs: the DL-based channel access (DL/CA+Iwl) and the DL-based MCS selection (CSMA/CA+DL/MCS). The purpose of this comparison is to evaluate whether our proposed joint functional design offers a significant performance advantage over individual functional design approaches.

As shown in Fig. 9a, DL-MAC shows higher throughput compared to both CSMA/CA+DL/MCS and DL/CA+IWL across all time intervals. This superior throughput performance demonstrates that the joint functional design more effectively optimizes channel access and rate adaptation, leading to improved data transmission efficiency. From Fig. 9b, DL-MAC achieves the lowest mean delay, which aligns with its higher throughput. In contrast, both individual functional design approaches incur higher mean delays, further supporting the notion that the joint design approach can achieve faster packet transmission. Overall, this comparison highlights the effectiveness of the joint design of channel access and MCS selection, suggesting potential synergistic benefits. When these functions are designed separately, they may not achieve optimal synergy, which may lead to potential limitations in overall system performance.

5.4.4. LSTM versus plain DNN

We investigated the impact of different neural network architectures on the performance of the joint design of channel access and rate adaptation within DL-MAC. Specifically, we examine 2 architectures: LSTM, as previously proposed, and a plain DNN. For the plain DNN, we just employ a straightforward fully connected network with 3 hidden layers, which consist of 512, 128, and 64 neurons, respectively. ReLU is employed as the activation function for the hidden neurons. Fig. 10 presents a comparison of these 2 schemes, focusing on their performance in terms of throughput and mean delay. The error bars represent 3 times the standard deviation, derived from the mean and variance obtained over 10 experiments.

As we can see from Fig. 10a, the LSTM model consistently achieves higher throughput across the measured time intervals than the plain DNN model. This is attributed to the LSTM's inherent suitability for processing time series data, enabling it to handle RSSI sequences more



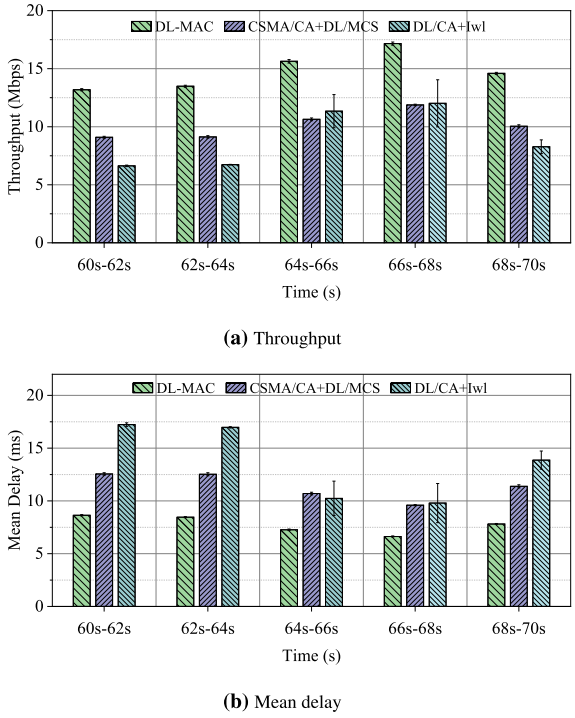
(a) Throughput on channel 1

(b) Mean delay on channel 1

(c) Throughput on channel 11

(d) Mean delay on channel 11

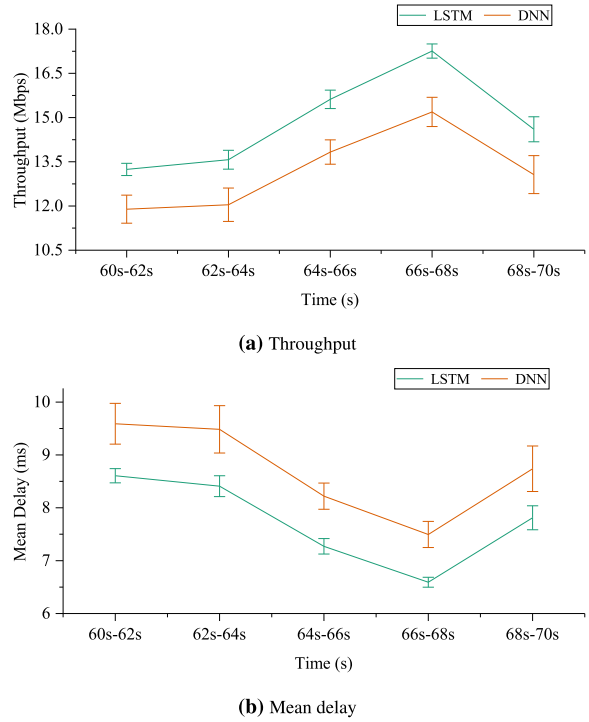
Fig. 8. Generalization performance comparisons of DL-MAC on channels 1 and 11: (a) Throughput on channel 1; (b) Mean delay on channel 1; (c) Throughput on channel 11; (d) Mean delay on channel 11.



(a) Throughput

(b) Mean delay

Fig. 9. Performance comparisons of joint vs. individual functional designs: (a) Throughput; (b) Mean delay.

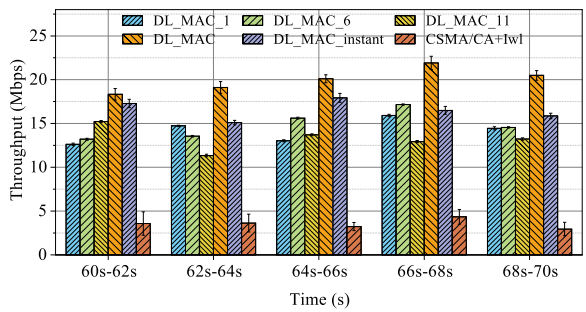


(a) Throughput

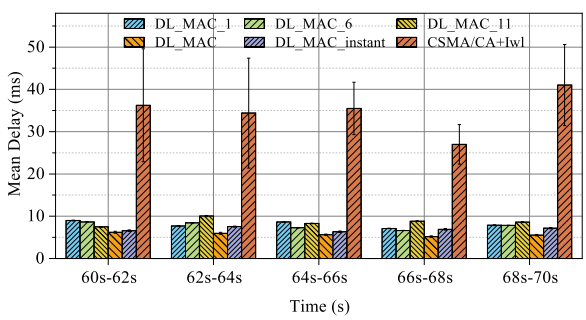
(b) Mean delay

Fig. 10. Comparing the performance of schemes using LSTM and plain DNN: (a) Throughput; (b) Mean delay.

effectively for DL-MAC applications. Moreover, as observed in Fig. 10b, the LSTM model displays a lower mean delay, indicating its potential for tasks that demand for both high throughput and low latency. On average, the LSTM model achieves approximately 12.5% higher throughput and an 11.2% reduction in mean delay compared to the plain DNN model, highlighting its superior ability to capture temporal dependencies in RSSI data. This enables the LSTM to more effectively optimize the task of joint channel access and rate adaptation based on a series of RSSI data during online testing.

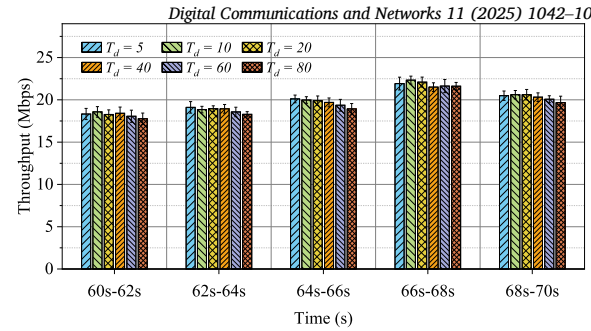


(a) Throughput

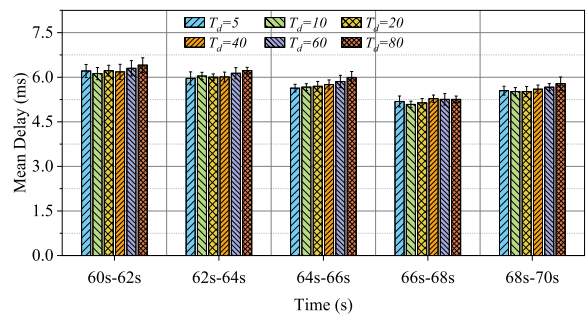


(b) Mean delay

Fig. 11. Evaluating channel switching performance of DL-MAC: (a) Throughput; (b) Mean delay.



(a) Throughput



(b) Mean delay

Fig. 12. Performance comparisons under different channel switching overheads (T_d): (a) Throughput; (b) Mean delay.

5.5. Performance evaluation across multiple channels

In this set of experiments, we designate channels 1, 6 and 11 as the available options for AI device operations. This configuration allows DL-MAC to dynamically select the optimal channel from these channels for access and rate adaptation at varying time intervals. For DNN training, we utilize data collected from these 3 channels during the first 60 seconds. We then partition the dataset into training and validation subsets, allocating 80% to training and the remaining 20% to validation. An early stopping mechanism is implemented to prevent overfitting during the offline training process. Once trained, the DNN is utilized online to perform the channel switching process. The performance of DL-MAC is then evaluated using a separate set of data from the following 10 seconds. Likewise, each experiment is replicated 10 times, and we report the mean and standard deviation for all assessed metrics. Likewise, the training process is conducted offline, and the testing process is conducted online.

5.5.1. Performance comparison of channel switching

Fig. 11 displays performance of DL-MAC across channels 1, 6 and 11. In the figure, the performance of DL-MAC operating on channels 1, 6 and 11 is denoted as DL-MAC_1, DL-MAC_6 and DL-MAC_11, respectively. Additionally, we have introduced an instant channel switching algorithm named DL-MAC_instant. This algorithm performs real-time predictions of all available channels immediately after each transmission, selecting the optimal channel for the next channel access and transmission. This achieves real-time optimization of channel selection. Furthermore, we have compared DL-MAC with the traditional CSMA/CA protocol for channel access combined with the Intel IWL algorithm for rate adaptation.

The results presented in Fig. 11 demonstrate the performance advantages of DL-MAC with channel switching capabilities. We have 3 main observations from Fig. 11:

- Our proposed DL-MAC protocol with channel switching capability demonstrates higher throughput and lower delay in multi-channel environments compared to when DL-MAC operates on a single channel. Despite the inherent overheads associated with channel switching, the overall performance improvement validates the effectiveness of the channel switching capability. The impact of channel switching overheads has been experimentally verified in subsection 5.5.2.
- The instant channel switching algorithm, although capable of rapidly adapting to changes in the network environment and achieving good throughput and delay performance, still falls short of DL-MAC with integrated channel switching capability. The instant channel switching algorithm can predict and select the best channel immediately after each transmission. However, the rapid changes in channel conditions and frequent switching ultimately lead to reduced throughput and increased latency in practical applications.
- For devices employing CSMA/CA for channel access coupled with the Intel IWL algorithm for rate adaptation, there is a significant reduction in throughput as well as an increase in mean delay. The primary reason for this performance degradation is the reliance of the IWL algorithm on historical information from previous channel conditions for rate decisions. Consequently, when a device transitions to a new channel, the lack of the real-time detection and evaluation of new channel conditions impacts the accuracy of rate adaptation. Often, the Intel IWL algorithm may select a rate that is below the optimal or feasible rate due to the lack of current channel data. This situation can severely degrade network performance, especially in environments with frequent channel switching.

5.5.2. Impact of channel switching overheads

We next examine the impact of channel switching overheads, denoted by T_d, on the performance of DL-MAC. Fig. 12 illustrates the performance comparison under varying channel switching overheads. As observed from Fig. 12a, the throughput remains relatively consistent across a range of T_d values, from $T_d = 5$ to $T_d = 80$. There is no significant reduction in throughput as T_d increases, suggesting that DL-MAC can effectively manage channel switching without a substantial impact

on throughput within the tested range of T_d. Similar to the throughput, the mean delay across different T_d values remains relatively consistent, without noticeable increases as T_d rises, as shown in Fig. 12b. This suggests that DL-MAC can maintain timely packet delivery even as the costs of channel switching vary.

In summary, the performance of DL-MAC in terms of throughput and mean delay is not significantly affected by the channel switching overheads within the tested range. This indicates that DL-MAC is effectively managing the channel switching process, maintaining performance despite the incurred delay from switching. The experimental results suggest that DL-MAC exhibits resilience to the overheads introduced by channel switching, which is a desirable feature in dynamic multi-channel environments.

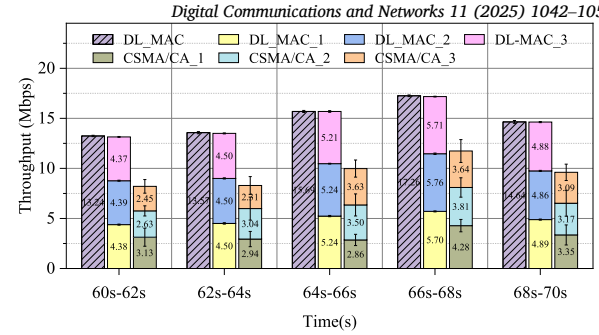
5.6. DL-MAC reformulation

Note that all of our previous experiments were conducted in an environment where only an intelligent device coexists with multiple legacy devices. However, when there are multiple intelligent devices in the DL-MAC network, the complexity of the system significantly increases due to all intelligent devices making independent decisions within the same environment. This is because each intelligent device can be considered an entity whose transmission behaviors need to be learned by the other intelligent devices.

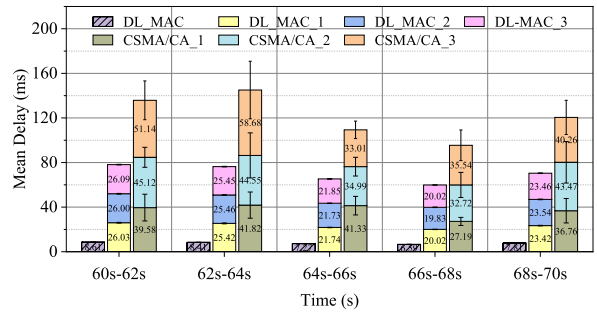
To address this issue, we have reformulated our system model as a semi-distributed system. Within this system, all intelligent devices are coordinated by a central DL-MAC gateway, akin to the Point Coordination Function (PCF) protocol in the IEEE 802.11 MAC architecture. In this model, the DL-MAC gateway is first subjected to offline training, a process consistent with that of a single intelligent DL-MAC device. Following the completion of the gateway's training, it can then be employed for online testing. In each time slot, the DL-MAC gateway decides whether intelligent devices within the DL-MAC network should access a channel. If the gateway decides to allow channel access and transmission, it selects an intelligent device to carry out the transmission via a polling mechanism. After transmission, the chosen intelligent device receives feedback from the system and communicates with the gateway, which then performs the corresponding RSSI compensation. If the gateway decides not to grant channel access, then all intelligent devices remain silent. In this manner, the DL-MAC gateway can essentially be regarded a large virtual proxy composed of multiple intelligent devices. As for the coordination information between the gateway and the intelligent devices it manages, this can be achieved through signal transmission over the control channel, which is typically completed within a short time slot. In this paper, we do not delve into this part in detail because our focus is primarily on the model architecture rather than the implementation specifics.

In our experimental setup, a DL-MAC gateway includes 3 intelligent devices, all of which operate in a saturated traffic mode, meaning each device always has packets to send. Each device is equipped with a buffer to store the packets awaiting transmission, and the arrival process of these packets follows a Poisson distribution, as described in subsection 5.1.3. Unless specifically stated otherwise, all experimental settings remain consistent.

Fig. 13 displays throughput and mean delay performance of the DL-MAC gateway and the 3 DL-MAC intelligent devices. We use the performance of the gateway as the benchmark, considering that in heterogeneous network scenarios, the gateway can be viewed as an independent DL-MAC device. For comparison to our proposed DL-MAC, we also include 3 devices that employ the traditional CSMA/CA protocol for channel access and the Intel IWL algorithm for rate adaptation. In Fig. 13, DL-MAC is represented by the gateway proxy, while DL-MAC_1, DL-MAC_2, and DL-MAC_3 are the 3 intelligent devices managed by the gateway, respectively; CSMA/CA_1, CSMA/CA_2, and CSMA/CA_3 represent the devices that adopt the CSMA/CA protocol for channel access and the Intel IWL algorithm for rate adaptation, respectively. From the



(a) Throughput



(b) Mean delay

Fig. 13. Performance comparisons of 3 DL-MAC devices in a unified protocol setting: (a) Throughput; (b) Mean delay.

results in Fig. 13a, it is evident that the combined throughput of the 3 intelligent devices employing a polling mechanism can achieve performance levels comparable to that of a DL-MAC gateway. The gateway not only ensures overall system throughput but also promotes fairness among its managed intelligent devices, providing each with an equal opportunity for transmission. In contrast, devices utilizing CSMA/CA exhibit lower total throughput, and there are significant differences in fairness between devices, failing to guarantee an equal channel access opportunity for each device. As in Fig. 13b, it is clear that the DL-MAC gateway exhibits the best performance in terms of mean delay. It can be inferred that having only a single DL-MAC device configured by the DL-MAC gateway allows this device to have the opportunity for channel access and transmission in every communication cycle, thereby achieving the lowest delay. The mean delay of the 3 DL-MAC devices managed by the gateway is approximately 3 times that of the single device case, a result directly attributed to the operation of the polling mechanism, where each device takes turns accessing the channel and transmitting. Even so, compared to devices utilizing CSMA/CA, DL-MAC devices still demonstrate a significant advantage in mean delay under the polling mechanism. Moreover, devices managed by the DL-MAC gateway maintain a uniform delay, with no device experiencing unreasonable delays, proving DL-MAC's fairness in delay management. For CSMA/CA devices, there is a notable variation in delay among devices. Therefore, the DL-MAC gateway and the managed devices exhibit significant advantages in ensuring high throughput, low mean delay, and maintaining fairness, further proving the effectiveness and reliability of DL-MAC in handling high-density network communications.

6. Conclusions

In this paper, we have proposed DL-MAC, a cross-layer decision-making strategy that incorporates deep learning techniques. This protocol integrates functionalities for channel switching and joint channel access and rate adaptation, aiming to optimize throughput and minimize packet delay in complex and heterogeneous networks. The DL-MAC framework utilizes a DNN for precise channel selection and employs

an RNN to efficiently manage joint channel access and rate adaptation on the selected channel. By integrating these functions into a unified system, DL-MAC achieves a multifunctional MAC protocol with a cross-layer design, significantly improving spectrum utilization and network performance. Experimental results on the datasets collected by our laboratory demonstrate that DL-MAC exhibits significant performance improvements in terms of throughput and latency compared to traditional MAC protocols, proving its effectiveness in enhancing wireless network performance.

In future work, we aim to investigate effective methodologies for characterizing spectrum data sampled from the real world to facilitate deeper data analysis. Future research should focus on developing new data processing and analysis techniques to handle and utilize large scales of real-world spectrum data, providing guidance for optimizing and improving the performance of wireless communication systems.

Another possible research direction involves designing more efficient algorithms to further optimize network throughput and minimize latency while considering fairness across the network. We will also investigate the joint design of channel access and transmission power control strategies. Currently, DL-MAC primarily focuses on channel access and rate adaptation. Incorporating power control into the considerations can effectively manage the strength of wireless signals, reducing interference, improving communication quality, and enhancing energy efficiency in the future.

CRediT authorship contribution statement

Jiantao Xin: Writing – original draft, Software, Methodology, Investigation, Data curation. **Wei Xu:** Validation, Methodology, Formal analysis, Conceptualization. **Bin Cao:** Supervision, Methodology, Investigation, Conceptualization. **Taotao Wang:** Writing – review & editing, Supervision, Methodology, Investigation, Funding acquisition, Conceptualization. **Shengli Zhang:** Supervision, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Bin Cao is an editorial board member/editor/editor-in-chief for Digital Communications and Networks and was not involved in the editorial review or the decision to publish this article. All authors declare that there are no competing interests.

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